Problem-Solution Fit

Date	15 March 2026
Team ID	xxxxxx
Project Name	Credit Card Auto-Approval Prediction System
Maximum Marks	5 Marks

1. CUSTOMER SEGMENT(S) [CS]

Target Audience:

- Retail banking credit card departments and operations managers.
- Risk assessment officers looking to expedite candidate screening.
- Individual loan/credit card applicants seeking preliminary eligibility feedback.

6. CUSTOMER LIMITATIONS
EG. BUDGET, DEVICES **[CL]**

Limitations:

- No specialized hardware (needs to run on standard browser/mobile interfaces).
- Limited budget for enterprise integration.
- Requires a simple, easy-to-understand layout with no complex setup.

5. AVAILABLE SOLUTIONS
PROS & CONS **[AS]**

Solution A:Manual Underwriting

- Pros: Personalized review.
- Cons: Extremely slow, high operational cost, subjective bias.

Solution B:Basic Rule-based Credit Scoring

- Pros: Fast calculation.
- Cons: Rigid, ignores non-linear relationships in demographic data, higher rate of incorrect rejection.

2. PROBLEMS / PAINS

+ ITS FREQUENCY **[PR]**

- Pains: Manual evaluation of credit card applications is slow (taking days/weeks), prone to human bias, and uses outdated rigid scoring frameworks.
- Frequency: Constant/Daily. Banks evaluate thousands of applications daily, leading to massive operational backlogs and customer dissatisfaction due to wait times.

9. PROBLEM ROOT / CAUSE [RC]

- Root Cause: Lack of automated, data-driven classification tools. Human underwriting cannot scale with high transaction volumes, and static scoring models do not effectively process complex correlations in applicant demographics (income levels, education, and employment history).

7. BEHAVIOR

+ ITS INTENSITY **[BE]**

- Behavior: Staff spend hours manually copying values across spreadsheets to cross-reference data.
- Intensity: High. The repetitive manual process directly causes fatigue, human error, and prolonged customer wait times.

3. TRIGGERS TO ACT [TR]

Triggers:

- High volume of application backlogs.
- Competitors offering "instant approval" options.
- Need to reduce administrative costs and improve risk assessment accuracy.

4. EMOTIONS

BEFORE / AFTER **[EM]**

- Before: Anxious, frustrated, and uncertain about application status and risk factors.
- After: Confident, satisfied, and informed with instantaneous, objective decisions and confidence scores.

10. YOUR SOLUTION [SL]

- CardApprove AI: A web-based, real-time prediction application powered by a Random Forest ML model (95.8% accuracy). The app takes demographic inputs (income type, annual income, employment, education) and outputs immediate approvals/rejections with confidence scores, eliminating wait times and manual operational bottlenecks.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Online: Web browsers, internal bank intranet portals, and mobile banking applications.

OFFLINE

- Offline: Physical paper-based credit applications filed at bank branches.